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Project Description Questions

1. Type of System: Real Estate System
2. System Purpose: Manage property details such as addresses, tenants, leases, maintenance requests, and customer feedback
3. Who will use the system:
    a. Property Managers
    b. Staff
    c. Customers/Tenants
4. Required Entities in the System:
    a. Employees
    b. Properties
    c. Tenants
    d. Leases
    e. Maintenance Requests
    f. Rent/Payments
    g. Customer Ratings/Feedback
5. What does each entity represent?
    a. Employees
        i. What it is: Person(s) working to maintain properties
        ii. Why it's needed: Without employees, there is no one to fulfill maintenance requests
        iii. Role in the system: Serves as a method to fulfil maintenance requests.
    b. Properties
        i. What it is: Addresses of the properties being leased
        ii. Why it's needed: Without properties, there is nothing to assign a tenant to
        iii. Role in the system: Serves as a base that a tenant can be applied/assigned to
    c. Tenants
        i. What it is: Person(s) leasing the properties
        ii. Why it's needed: Customers of the system. They make a property "active" and allow services to be provided
        iii. Role in the system: Serve as the customers for the system
    d. Leases:
        i. What it is: Contract of the tenant
        ii. Why it's needed: Defines the rules of the tenant at a property
        iii. Role in the system: Sets the start and end dates of the residence, defines rent cost
    e. Maintenance Requests
        i. What it is: Requests from the customer for maintenance at their property
        ii. Why it's needed: Ensures the upkeep of the properties

iii. Role in the system: Provides a method of communication between staff and customer

f. Rent/Payments
   i. What it is: Payment for rent of the tenant
   ii. Why it's needed: Method of business and income for the manager and staff
   iii. Role in the system: Identifies if a customer has paid and if that payment is on time

g. Customer Rating
   i. What it is: Customer feedback about properties, management, etc
   ii. Why it's needed: Allows a customer to express level of satisfaction with service
   iii. Role in the system: Serves as a method of communication between customer and staff

6. What information should the system store about each entity?
   a. Employees
      i. Title/Position
      ii. Email
      iii. Phone number
      iv. Hours
      v. Pay
   b. Properties
      i. Address
      ii. Status
      iii. Price
      iv. Number of bedrooms
      v. Number of bathrooms
   c. Tenants
      i. Name
      ii. Email
      iii. Phone Number
      iv. Address
   d. Leases
      i. Landlords name
      ii. Landlord email
      iii. Tenantes name
      iv. Tenantes email
      v. Other occupants
      vi. Address
      vii. Term
      viii. Security Deposit
      ix. Cost
   e. Maintenance Requests
      i. Address

        ii. Date Request was placed
        iii. Status
        iv. Issue/Note
    f. Payments
        i. Address
        ii. Date due
        iii. Date payed
        iv. Customer name
    g. Customer Rating
        i. Name
        ii. Stars 1-5
        iii. Note
7. How are the entities connected?
    a. A manager can add a property and assign a tenant with particular lease details
    b. A customer can submit a maintenance request that can be viewed and addressed by managers and employees
    c. A customer can submit a feedback rating that can be viewed by managers
    d. An employee can view a report of any open maintenance requests submitted by customers
    e. A manager can see if a customer has submitted a rent payment on time
8. What should users be able to do in the system?
    a. Managers
        i. Add properties
        ii. Add tenants
        iii. View payment status
        iv. View open maintenance requests
        v. View customer ratings/feedback
    b. Staff
        i. View tenants and their assigned address
        ii. View submitted maintenance requests
        iii. Close maintenance requests once completed
    c. Customers/Tenants
        i. Submit rent payments
        ii. Submit maintenance requests
        iii. Submit Management feedback/ratings
9. What rules should the system follow?
    a. A customer cannot change their lease details
    b. Once submitted by a customer, only a manager or maintenance staff can close a request
    c. Only a manager can add or remove properties and tenants
    d. A tenant must be applied to a property
    e. A tenant can only be applied to a vacant property
    f. A rent value must be assigned to a tenant and property

10. What is one example of how someone would use the system?
    If a tenant notices that their faucet is broken, they can send in a request to get it fixed.
11. What extra entities could make the system better?
    a. Payment Method
    b. Late fees
    c. Inbox
    d. Notifications
    e. Schedule to view properties
12. What should the final project description explain?
    a. Our team selected a Real Estate System. The purpose of this system is to manage property details such as addresses, employees, tenants, leases, maintenance requests, and customer feedback. The main users of this system are property managers, employees, and tenants.
    b. The required entities in our system are employees, properties, tenants, leases, maintenance requests, and customer feedback. These entities are important because they all work together. Without employees, there is no one to fulfill maintenance requests, without maintenance requests, there's no easy way to maintain your property.
    c. The system will store information such as the tenants names, contact information, and addresses. It will also store information about the properties, how many bedrooms and bathrooms they have, and the price. The entities are connected because a manager can add a property and assign a tenant with particular lease details. A customer can submit a maintenance request that can be viewed and addressed by managers and employees.
    d. The system will allow users, specifically tenants to submit rent payments, submit maintenance requests, and submit Management feedback/ratings. For example, if a tenant notices their faucet is broken they can log into the system and send in a request to get it fixed. We may also include additional entities such as payment methods, late fees, inbox, notifications, and a way to schedule to view properties. To make the system more complete.

Final Project Description

Our team selected a Real Estate Management System. The purpose of this system is to manage properties and the tenants leasing them. The main users of the system are Managers/Landlords, Employees (Maintenance Staff), and the Tenants.

The required entities in our system are Properties, Employees, Tenants, Leases, Maintenance Requests, Payments, and Customer Feedback. These entities are important because each plays their own role. For example, properties allow tenants a location to lease, employees are able to address maintenance requests submitted by the tenants, managers are able to set lease terms and monitor the timeliness of payments by the customers, and customers are able to submit feedback about the management of their leased property.

The system will store information such as names, titles, emails, hours, pay of employees, name, email, phone number and address of tenants, property details such as address, size, cost, and status whether occupied or vacant, lease details including the names and emails of landlords and tenants, whether there are other occupants, which property the lease is for, the duration, deposit, and cost associated, the address and request date of maintenance requests, along with their status and issue to be addressed, address, due date, date paid, and customer name of lease payments, and the name, rating, and notes of customer feedback. The entities are connected because each has ways of interacting with the users. For example, landlords can add or remove properties and assign tenants to vacant properties, customers can view their lease details and make rent payments or submit requests for maintenance that can be seen and addressed by maintenance staff, and landlords can view feedback submitted by their tenants.

The system will allow tenants to view the property, submit reviews, make requests, make payments, and view their leases. Employees get access to a few more functions as they can view the tenants address information, view each repair request, and close the request when the task is done. The managers have the most features as they can add or drop properties, view payment status, and add tenants. For example, if the manager wants to check if all his tenants have paid, he could double-check that. We may also include additional entities such as payment method details for rent payments, late fees that would be added to overdue rent payments, an inbox and notifications for managers and employees when updates occur, and a way for potential tenants to schedule property viewings to make the system more complete.